L14 — Reinforcement Learning

MDPs, policy optimization, and behavior synthesis

Study summary generated from lecture slides. ETH AI for Digital Characters.

ML paradigms

- Supervised: (data, label) pairs, backpropagation
- Unsupervised: learn latent representations (autoencoders)
- RL: agent-environment interaction, reward signal (no labels)
- RL for non-differentiable simulators (physics, game engines)

MDP and key concepts

- State s, action a, reward r, policy pi(a|s), return R (discounted sum)
- Trajectory τ: sequence of (s, a, r) from one episode
- Markov property: future depends only on current state

Value functions

- Q(s,a): expected return from taking action a in state s
- V(s): E_{a~pi}[Q(s,a)]
- Optimal policy maximizes expected return (not each individual reward)

Algorithms

- Policy gradient: optimize pi directly via sampled trajectories
- Actor-Critic: critic baseline reduces variance
- PPO: clipped surrogate objective, on-policy rollouts
- Q-learning: implicit policy = argmax_a Q(s,a); off-policy with replay buffer
- Reward does NOT need to be differentiable

PPO advantages

- Critic baseline reduces variance
- Clipped surrogate prevents destructively large policy updates
- Multiple epochs on same batch improves sample efficiency

Behavior synthesis applications

- Humanoid locomotion and goal reaching
- Soccer dribbling, motion tracking in simulation
- Formulate: state, action, reward, policy for each task
- Prior knowledge (e.g. z ~ N(0,I)) can be encoded as KL penalty